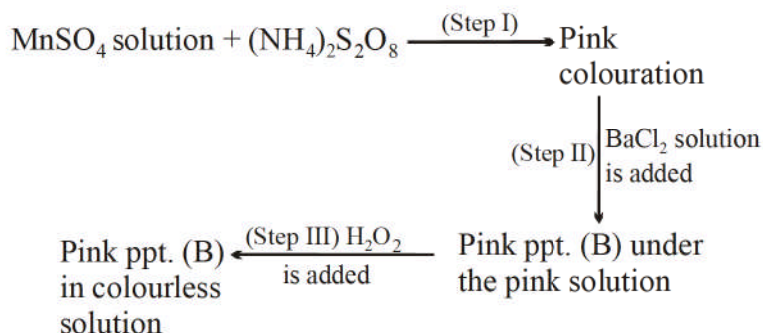


WORK SHEET - 09

MISCELLANEOUS

Comprehension

(Question No. 1 to 3)



- Q.1 The oxidation states of S in $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and in the corresponding pink precipitate are respectively.
(A) +6, +6 (B) +6, +4 (C) +7, +6 (D) +7, +4
- Q.2 The pink precipitate is of
(A) MnSO_4 (B) BaSO_4 (C) BaSO_3 (D) $\text{MnO}(\text{OH})_2$
- Q.3 Which of the following statements is correct?
(A) The oxygen atoms in peroxy linkage of $\text{S}_2\text{O}_8^{2-}$ are reduced
(B) The pink colour is due to adsorbed MnO_4^- on BaSO_4 surface
(C) H_2O_2 cannot reduce adsorbed MnO_4^- but can reduce MnO_4^- in solution
(D) All are correct

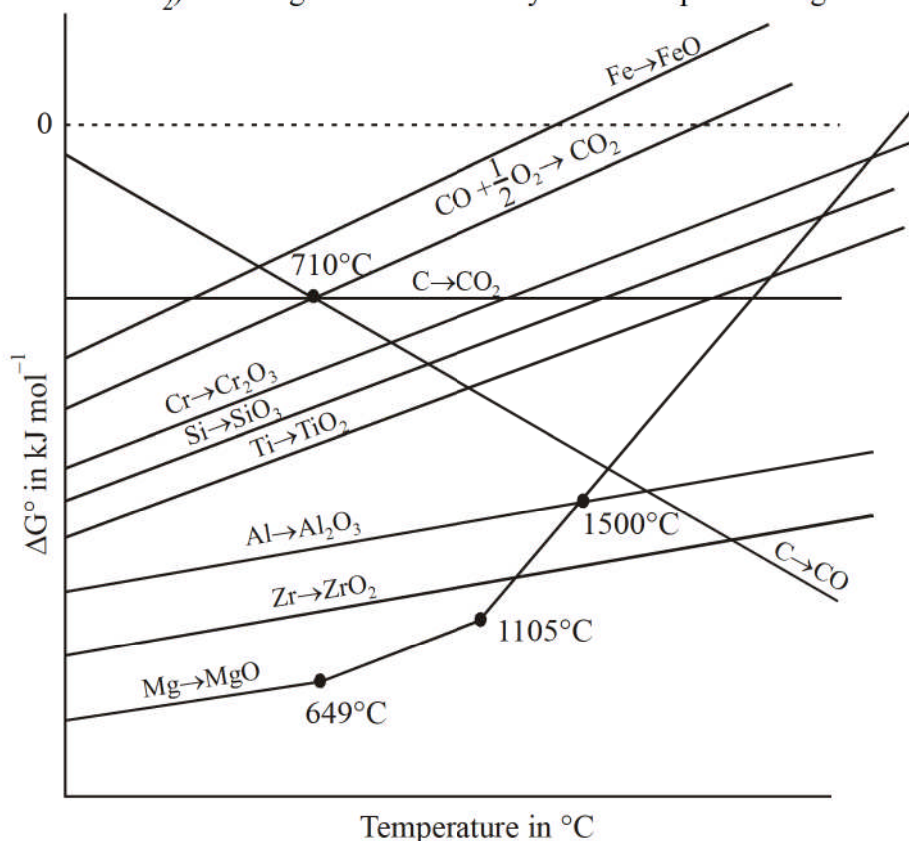
(Question No. 4 to 6)

There are two ores A_1 and A_2 of metal M. When ore A_1 is calcinated, a black solid S is obtained along with the liberation of CO_2 and water. The ore A_1 on treatment with HCl and KI gives a ppt. P and I_2 is liberated. The other ore A_2 on roasting gives a gas G and metal M is set free. When gas G is passed through $\text{K}_2\text{Cr}_2\text{O}_7$, it turns green.

- Q.4 The ore A_2 is
(A) malachite (B) azurite (C) ruby copper (D) copper glance
- Q.5 The colour of precipitate P is
(A) brown (B) white (C) yellow (D) green
- Q.6 Which of the following gases shows a similar behaviour towards $\text{K}_2\text{Cr}_2\text{O}_7$ as gas G?
(A) H_2S (B) SO_3 (C) CO_2 (D) SO_2

(Question No. 7 to 9)

The Ellingham diagram (Figure) represents the formation of oxides of several metals together with oxides of C (i.e. CO and CO₂). The diagram is not to scale yet it can help in making a lot of predictions.

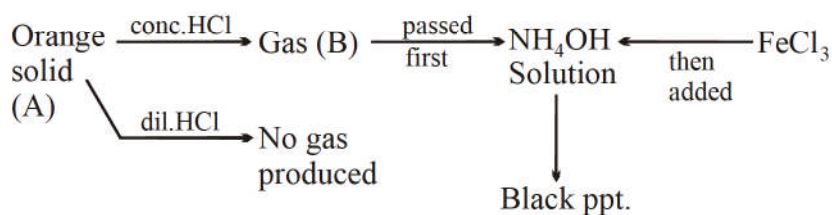


- Q.7 Which is the chief reducing agent if the carbon reduction of iron oxides is carried out below 710°C ?
- (A) CO
(B) C
(C) Either C or CO
(D) Cannot be predicted on the basis of given information

- Q.8 At 1100°C , which of the following reactions is thermodynamically most favourable?
- (A) $\text{TiO}_2 + \text{C} \longrightarrow \text{Ti} + \text{CO}_2$
(B) $\text{CO}_2 + \text{C} \longrightarrow 2\text{CO}$
(C) $\text{Al}_2\text{O}_3 + 3\text{C} \longrightarrow 2\text{Al} + 3\text{CO}$
(D) $\text{Cr}_2\text{O}_3 + 3\text{C} \longrightarrow 2\text{Cr} + 3\text{CO}$

- Q.9 Which of following statements is incorrect?
- (A) Si cannot reduce MgO at all.
(B) Al can reduce MgO at above 1500°C .
(C) Al can reduce Cr_2O_3 as well as TiO_2
(D) Al cannot reduce ZrO_2 but Mg can reduce ZrO_2 .

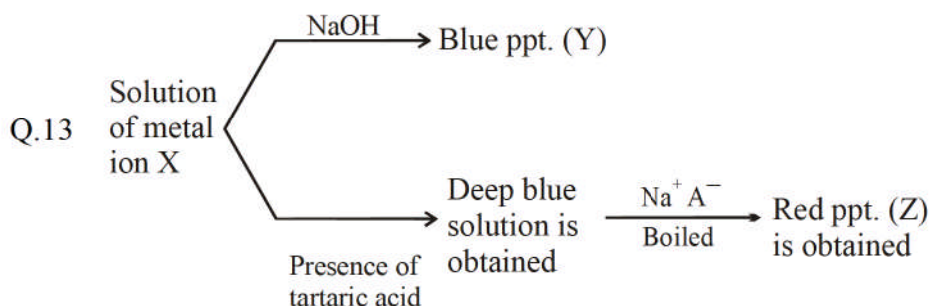
(Question No. 10 to 12)



- Q.10 The black precipitate is
- (A) FeS
(B) Fe_2S_3
(C) $(\text{FeS} + \text{S})$
(D) $(\text{FeS} + \text{Fe}_2\text{S}_3)$

- Q.11 The orange solid is
 (A) As_2S_3 (B) Sb_2S_3 (C) Sb_2O_3 (D) SbOCl
- Q.12 The metal(s) which reacts with gas B under hot condition to produce H_2 is (are)
 (A) Na (B) Cu (C) Pb (D) All of the above

Single correct



The cation X and anion A are respectively.

- (A) NH_4^+ , CH_3CO_2^- (B) HCO_2^- , Cu^{2+}
 (C) Pb^{2+} , $\text{C}_2\text{O}_4^{2-}$ (D) Cu^{2+} , HCO_2^-
- Q.14 NiCl_2 solution + NH_3 solution $\longrightarrow$ Green ppt. $\xrightarrow[\text{NH}_3 \text{ solution}]{\text{excess}}$ Deep blue solution

Finally Ni^{2+} ion is present in the

- (A) cationic part of the solution
 (B) anionic part of the solution
 (C) both in the cationic and anionic parts of solution
 (D) Can't predicted
- Q.15 The number of ores in which Pb is present is Galena, limonite, chalcocite, azurite _____.
 (A) 1 (B) 2 (C) 3 (D) 4

- Q.16 N_2H_4 reacts with conc. H_2SO_4 to produce a salt $[\text{NH}_3 - \text{NH}_3]^{2+} \text{SO}_4^{2-}$ in which
 (A) $d_{\text{N-N}}(\text{salt}) > d_{\text{N-N}}(\text{N}_2\text{H}_4)$ (B) $d_{\text{N-N}}(\text{salt}) < d_{\text{N-N}}(\text{N}_2\text{H}_4)$
 (C) $d_{\text{N-N}}(\text{salt}) = d_{\text{N-N}}(\text{N}_2\text{H}_4)$ (D) Cannot be predicted

- Q.17 Choose incorrect option:
 (A) Interstitial compounds have high melting points, higher than those of pure metals.
 (B) Interstitial compounds are very hard, some borides approach diamond in hardness
 (C) Interstitial compounds retain metallic conductivity
 (D) Interstitial compounds are chemically inert.
- Q.18 Which of the following statements are true:
 (A) In $[\text{PtCl}_2(\text{NH}_3)_2]^{2+}$ the cis form is optically inactive while trans form is optically active
 (B) In $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, geometrical isomerism does not exist while optical isomerism exists
 (C) In Mabcd square planar complexes, show both optical as well as geometrical isomerism
 (D) In Mabcd tetrahedral complex, optical isomerism can not be observed.

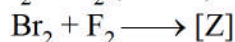
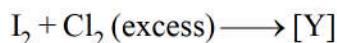
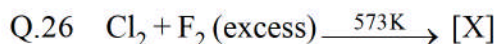
- Q.19 The number of two centre, two electron bonds in a molecule of $\text{Al}_2(\text{CH}_3)_6$ is :
 (A) 18 (B) 16 (C) 22 (D) 24

Assertion Reason

- Q.20 **Statement-1:** Most of the wet test observations of K^+ and NH_4^+ are identical.
Statement-2: Radius of K^+ ions is almost identical to that of NH_4^+ .
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
- Q.21 **Statement-1:** $SnCl_2$ and $HgCl_2$ cannot exist together.
Statement-2: $Sn^{2+} + Hg^{2+} \longrightarrow Hg \downarrow + Sn^{4+}$: occurs spontaneously.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

More than one may be correct

- Q.22 To identify I_2 produced in a particular reaction, the suitable reagents are
(A) excess NaOH solution (B) presence of CCl_4
(C) excess KI solution (D) presence of starch
- Q.23 Sometimes turbidity appears in the group II test of group analysis, even in the absence of group II cations. This is due to
(A) the presence of Fe^{2+} cation (B) the presence of Fe^{3+} cation
(C) the presence of Zn^{2+} cation (D) the presence of an oxidising anion
- Q.24 N_2O_4 reacts with NaOH to produce a colourless solution. Which are the correct statements among the following for this solution?
(A) It can give the brown ring test for nitrate
(B) It reacts with $AgNO_3$ to give white ppt.
(C) It decolourises $KMnO_4$ solution
(D) After treatment with $AgNO_3$, ppt. is filtered and filtrate is treated with $Zn + AcOH$, and the resulting solution does not respond towards Grise-Ilsvay test.
- Q.25 A greenish yellow gas with pungent and suffocating odour. It is about 2.5 times heavier than air when it reacts with
 $Al + [X] \longrightarrow [P]$
 $Na + [X] \longrightarrow [Q]$
 $Fe + [X] \longrightarrow [R]$
then choose the correct statements:
(A) [P] can form dimer
(B) Co-ordination number of [Q] is 6
(C) [R] is form neutral solution
(D) When [X] react sulphur it forms [Y] compound which has open book structure.



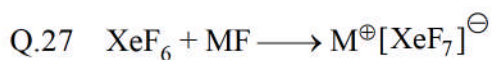
Choose the correct statements.

(A) [X] and [Y] both have same structure

(B) [Y] and [Z] are planar

(C) [X], [Y] and [Z] are polar

(D) [Y] is planar while [Z] is distorted octahedral geometry.



then M can be

(A) Na

(B) K

(C) Li

(D) Cs

Q.28 Which of the following metal is / are formed mixed oxides.

(A) Cr

(B) Mn

(C) Fe

(D) Co

Q.29 Which of the following are interstitial compounds.

(A) Mn_4N

(B) Fe_3H

(C) $\text{VH}_{0.56}$

(D) $\text{TiH}_{1.7}$

Q.30 Which of the following statements is/are correct?

(A) In $\text{Fe}_2(\text{CO})_9$, all CO bond length are not equal

(B) In $\text{Fe}_2(\text{CO})_9$, 3 CO bond length are larger rest of six are smaller than these three.

(C) Suppose bond length of CO is $1.128\text{\AA}$ and In $\text{Fe}_2(\text{CO})_9$, all CO bond length are larger than $1.128\text{\AA}$.

(D) $\text{Fe}_2(\text{CO})_9$ is an organo metallic compound.

Match the column

Q.31 Match the compound with the change observed on qualitative analysis.

Column I

(A) $(\text{NH}_4)_2\text{CO}_3$

(B) $\text{Ag}_2\text{C}_2\text{O}_4$

(C) $\text{Pb}(\text{NO}_3)_2$

(D) NH_4NO_3

(E) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

Column II

(P) Forms brown ring on reaction with FeSO_4 and conc. H_2SO_4

(Q) Leaves no residue on heating

(R) Colourless acidic gas is evolved on heating

(S) Residue is obtained on heating.

Q.32 Match the compounds with their common names and uses.

Column I

(A) $\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$

(B) $(\text{NH}_4)_2\text{SnCl}_6$

(C) SnS_2

(D) $\text{CaCN}_2 + \text{C}$

Column II

(P) Mosaic gold

(Q) Nitrolim

(R) Butter of tin

(S) Pink salt

Column III

(J) Used as fertilizer

(K) Used in calico

(L) Used in gilding purpose

(M) Used as mordant in dye

Integer type

Q.33 How many number of transition metal atoms have standard electrode potential (M^{+2}/M) less than copper.

Sc, Mn, Fe, Co, Zn, Cr, Ni

Q.34 Find the total number of compounds or ions having O—O bond longer than that in free O_2 molecule.

Na_2O_2 , KO_2 , $[Cr(O_2)_4]^{3-}$, $[CrO(O_2)_2(H_2O)]$, $[CrO_5(Py)]$, $[Cr(O_2)(CN)_5]^{3-}$, Na_2ZnO_2

Q.35 Find the number of complexes involving d^2sp^3 hybridisation of the central metal atom/ion.

$[Co(Ox)_3]^{3-}$, $[Mn(CN)_6]^{4-}$, $[Mn(NH_3)_6]^{2+}$, $[CoF_6]^{3-}$, $[Co(gly)_3]$,

$[Pt(NH_3)_6]^{4+}$, $[Ni(H_2O)_6]^{2+}$, $[Co(H_2O)_6]^{3+}$, $[IrCl_6]^{3-}$

Rankers Academy JEE

ANSWER KEY

Q.1	A	Q.2	B	Q.3	D	Q.4	D	Q.5	B
Q.6	A	Q.7	A	Q.8	B	Q.9	A	Q.10	C
Q.11	B	Q.12	D	Q.13	D	Q.14	A	Q.15	A
Q.16	B	Q.17	D	Q.18	B	Q.19	C	Q.20	A
Q.21	A	Q.22	BCD	Q.23	BD	Q.24	BC		
Q.25	ABCD	Q.26	ABCD	Q.27	ABD	Q.28	BCD		
Q.29	ABCD	Q.30	ABCD	Q.31	(A) QR (B) RS (C) S (D) PQ (E) S				
Q.32	(A) RM (B) SK (C) PL (D) QJ	Q.33	6	Q.34	6	Q.35	6		

